Final Report Guidelines and Template

Many organizations that employ engineers have standardized project progress reports. This is done to facilitate communication within the organization and to provide better control of work quality. Standardization permits routine checking of work as well as the opportunity to switch people to the tasks that are the most urgent.

An engineering report is usually written to tell someone (fellow engineers, supervisor, etc.) about work you have done. Frequently, you will have been given a problem and will be responding with your answer or solution to the questions raised by the request. You should give your recommended solution, supporting evidence, and your opinion of the solution, at the least. You may want to tell other things as well, but there is no reason to load up the report with unnecessary information from standard texts or minute detail that is of no value to the reader.

You will be required to electronically submit your Final Lab Report by 5pm on the day before your next prelab (Monday for Tuesday lab period, Wednesday for Thursday lab period). Submit as a Word document and use the following filename format:

FinalLab#TimeDayTeam#.docx

For example: FinalLab1AMTuesdayTeam3.docx

(Title Page)

Viscosity of Newtonian Fluids

Final Lab Report
Unit Operations Lab 1
September 1, 2009

Group 1
Name 1
Name 2
…

Basically, your **Final Lab Report** should tell what you did, why you did it, what you learned and the significance of your results. The report does not have to be extremely long. In fact, the key to writing an excellent report lies in presenting concisely your response to the objectives of the project and justifying that response with appropriate documentation. The **Final Repor**t should consist of the following sections:

1. Executive Summary (Examples at end of the Guideline) (10 pts)

The executive summary is the most important section of any report. It should be designed for your intended audience so that they can decide whether it is relevant for them to spend their time reading or not. Therefore, the executive summary should contain three parts: an objective, a brief statement of your results, and a brief statement of your overall conclusions. The objective should remind your boss of why the experiment was done because they are supervising many projects, and so cannot remember the details of each project. Remember that your audience is a hypothetical supervisor. Therefore, do not write the objective as if you were writing the report to your professor. Do not say things like: "This experiment was done to learn more about blah blah blah." Instead, write something like: "The objective of this experiment was to determine the heat transfer coefficient for an agitated vessel for use in the design of similar heat transfer vessels." The Executive Summary should then contain a brief overview of your experimental results. It should not contain a list of all of your observations. Instead, you should include only a brief paragraph of a few sentences discussing what you found. For example: "The heat transfer coefficient for an agitated vessel was found at three agitator speeds." This is brief, and tells the reader that they do not have to read this report if they are not interested in heat transfer coefficients. Next include specific results that will The results are factual statements of what you did. You have not yet stated whether the experiment succeeded or not. You should now include a brief paragraph stating your conclusions. Conclusions are your opinions concerning the facts based on your engineering judgement. For example, if your data did not satisfy the expected correlation, you should say so. Also, your conclusions should be directly related to your stated objective. If your objective was to test a given correlation, then your conclusion should state whether you were successful or not.
As a guide, the Executive Summary should contain approximately 200-300 words for laboratory reports such as these.

2. Introduction (10 pts)
Experimental objective, significance of the experiment,brief description of the apparatus and experiments, industrial applications

3. Theory (10 pts)
Describe the theory and principles related to the experiment and its objectives. Briefly describe how the experimental data will be used to obtain your results. Include important equations or empirical correlations.

4. Results (15 pts)

The Results section delivers the evidence that will help answer the questions raised by the objectives of the investigation, should prepare readers for the more detailed upcoming discussion, and justify the conclusions that will be drawn next. This should be the most extensive portion of your report; concentrate on making this section informative and readable. **Remember that results are the numbers you measured in the lab or calculations based on these numbers**. State the most important results first. Observations, data and calculated results (in consistent units) are often presented best as graphs or charts, particularly if it will be important to illustrate trends. However, tables make sense when you need to present accurate data and specific facts or demonstrate the relationships between numerical and/or descriptive data. Figures and tables should include – whenever possible – published, theoretical, and/or model/simulation values available from the literature or produced programmatically.

Tables of raw experimental data are not placed in this section; they are reported in Appendix I. Calibration curves and other results that do not relate directly to the upcoming discussion can either be reported in the Appendix, Data are often summarized or *reduced* for presentation. Reducing the data allows generalizations to be made and trends to be pointed out.

Charts and tables must be accompanied by a discussion of its significance in the body of the report. Every figure or table itself is numbered and supplied with a brief but descriptive title or caption. (Figure 1 Enthalpy vs Temperature, Table 1 Heat transfer coefficients at various RPM). Use clear X and Y axis titles, with units.

The Results section text briefly explains how the results were obtained from the experimental data, the associated quantitative uncertainty (e.g., confidence limits, standard error, etc.), references to appropriate equations or sample calculations, and any critical assumptions or approximations made in obtaining the results.

5. Discussion (15 pts)

In the Discussion section, critically analyze the results, both their magnitudes and trends, and compare them with literature values. Theory should always provide the underlying basis for this analysis. Provide sufficient detail to support all of your conclusions in the next report section. There are many questions that could be answered in this section so you should not limit yourself to those offered here as examples. Answer the questions that make most sense for your work.

- Do the results agree with theory, with the work of others, with models/simulation? If so, how? If not, why not? Can the disagreement be explained?
- What are the most probable sources of experimental error and have these affected your ability to draw conclusions? How might these errors have been reduced or eliminated?
- Did your results reveal problems with the experimental plan, method, or equipment? How might these be improved?
- Were your assumptions suspect or reliable?

- What definite conclusions can you draw from your results? What conclusions are more speculative?
- What implications do your results have on current practice or theory?
- What questions remain unanswered? What questions should be answered next?

6. References (10 pts)

Begin your References section here. Include a page break at the end of this section to start your Appendices.

7. Appendix I: Data Tabulation/Graphs (10 pts)

Begin your Data Tabulation/Graphs section here.

8. Appendix II: Error Analysis (10 pts)

Begin your Error Analysis section here.

9. Appendix III: Sample Calculations (10 pts)

Begin your Sample Calculations section here.

10. Appendix IV: Individual Team Contributions

See template in Blackboard

FURTHER DETAILS ON EXECUTIVE SUMMARIES

(Excerpt from unknown source, CHE Department, Stanford University)

An Executive Summary is a condensation of important information from a report or technical summary. It is usually one paragraph in length (approximately 200 words). It contains no equations, figures, tables, graphs, or references. It should be able to stand on its own, without either the title or the text, and should contain key words for referencing. An Executive Summary should be written so that the maximum amount of information is contained in the minimum amount of space.

The Executive Summary should be written after the report is complete. It should be able to answer what was done, why it was done, how it was done, the outcome, and the significance of the results. Several general rules about abstracts are as follows.

1) Delete any sentence that does not make a positive contribution.
2) The first sentence should say what was done.
3) Address the approach of the experiment.
4) State the most important results.
5) State conclusions and significance

The first sentence should inform the reader of what was done, and sometimes why it was done. Examples of first sentences could be:

- A materials balance approach was used to determine the location of the flame front for an *in-situ* oil shale retort.
- The drag on a sphere in a viscous fluid was measured as a function of sphere size and flow rate.
- The diffusivity of bovine serum albumin (BSA) in K-quark electrophoretic gel was measure and compared to literature values for BSA diffusivity in the major competing gels.

The next sentence or two should describe the method that was used to make the evaluations or comparisons raised in the first sentence. For example:

- Off gases from an Occidental Oil (OXY-9) oil shale retort were analyzed by gas chromatography. The resultant values were used in a computer model that calculated the mass of oil shale combusted as a function of off-gas production and the location of the flame front.
- Sphere of various sizes were attached to an apparatus designed to measure torque and placed in a channel of fluid that could be pumped at different rates. The torque measurements were converted into drag for each of the experiments and compared to predictions calculated according to the Sangani method.
- Concentrated radiolabeled BSA was applied to one side of the gel. Strips of the gel were removed at hourly intervals. Radiolabel content was analyzed as a function of distance from the source. The diffusivity was calculated from this information based on a one-dimensional model.

The next part of the Executive Summary states the results. See the examples below:

- The flame front predicted by the off-gas analysis lagged that indicated by the buried thermocouples by 15%.
- The experimental results agreed well with the Sangani calculations at low flow rates (Re < 1) but deviated significantly at higher flow rates (Re > 10).
- The measured diffusivity of BSA in K-quark gel, 5.2×10^{-5} cm/s, was less than half that of the competing gels.

Finally, the significance of the results is stated:

- Later mine analysis showed this to be the result of significant tunneling of the flame front. Both in-place thermocouples and off-gas analysis gave useful information about the location of a flame from in an *in-situ* oil shale retort.
- The Sangani approach is quite effective for the prediction of the drag on a sphere for creeping flow conditions but must be used carefully at higher Reynold's number flow.
- Thus the K-quark gels will give better separation of components for lengthy separations than will the competing gels.

REFERENCES for this Report Template Document:

1. Final Report Template Fall 2009, Department of Chemical Engineering, University of Illinois at Chicago, Chicago, IL.
2. CHE4162 Syllabus Spring 2010, Department of Chemical Engineering, Louisiana State University, Baton Rouge, LA.